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AI-Based Job Offer Risk Assessment and Scam Detection System

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Guide:

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INTRODUCTION

- Online job and internship scams are increasing rapidly
- Students and job seekers are the main targets
- Fake job offers look attractive and convincing
- Scammers promise high salary and easy work
- This project helps users identify scam job offers using AI
- Many victims lose money, personal data, and confidence
- Manual verification is difficult and time-consuming
- There is a strong need for an automated AI-based scam detection system

ABSTRACT

Online job and internship scams have become a serious problem, especially for students and fresh graduates. Fraudsters often create fake job offers that promise high salaries, easy work, and quick hiring processes. Many users fall victim to these scams due to lack of awareness and proper verification mechanisms.

This project proposes an **AI-Based Job Offer Risk Assessment and Scam Detection System** that helps users identify whether a job offer is genuine or fraudulent. The system analyzes various job-related details such as job title, company name, salary offered, job description, contact email, and job application URL. Natural Language Processing (NLP) techniques are used to analyze the text content of job messages, while rule-based checks are applied to detect common scam indicators like payment requests, unrealistic salaries, free email domains, and suspicious links.

EXISTING SYSTEM

- Job offers are verified manually by users
- Users depend on internet search or friends
- Some systems check only message text
- No clear risk score or explanation • Many fake job offers remain undetected

Disadvantages:

- Cannot verify companies in real time
- Accuracy depends on training data
- Internet-based scams keep evolving

PROPOSED SYSTEM

- AI-based job scam detection system
- Uses NLP and rule-based analysis
- Checks salary, payment, email, and URLs
- Generates risk score and risk level
- Displays results with graphs and explanations

Advantages:

- Easy-to-use and beginner-friendly interface
- No technical knowledge required
- Fast and accurate analysis
- Provides explainable AI results
- Detects both job and internship scams

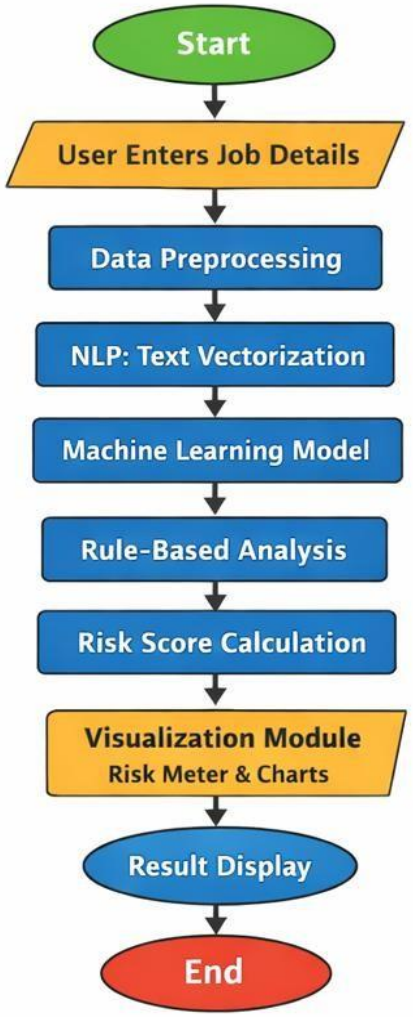
TECHNOLOGIES USED

- **Python** – Main programming language
- **Streamlit** – Web application framework
- **Scikit-learn** – Machine learning library
- **Natural Language Processing (NLP)** – Text analysis
- **Pandas & NumPy** – Data processing
- **Matplotlib** – Graphs and visualization

ALGORITHMS USED:

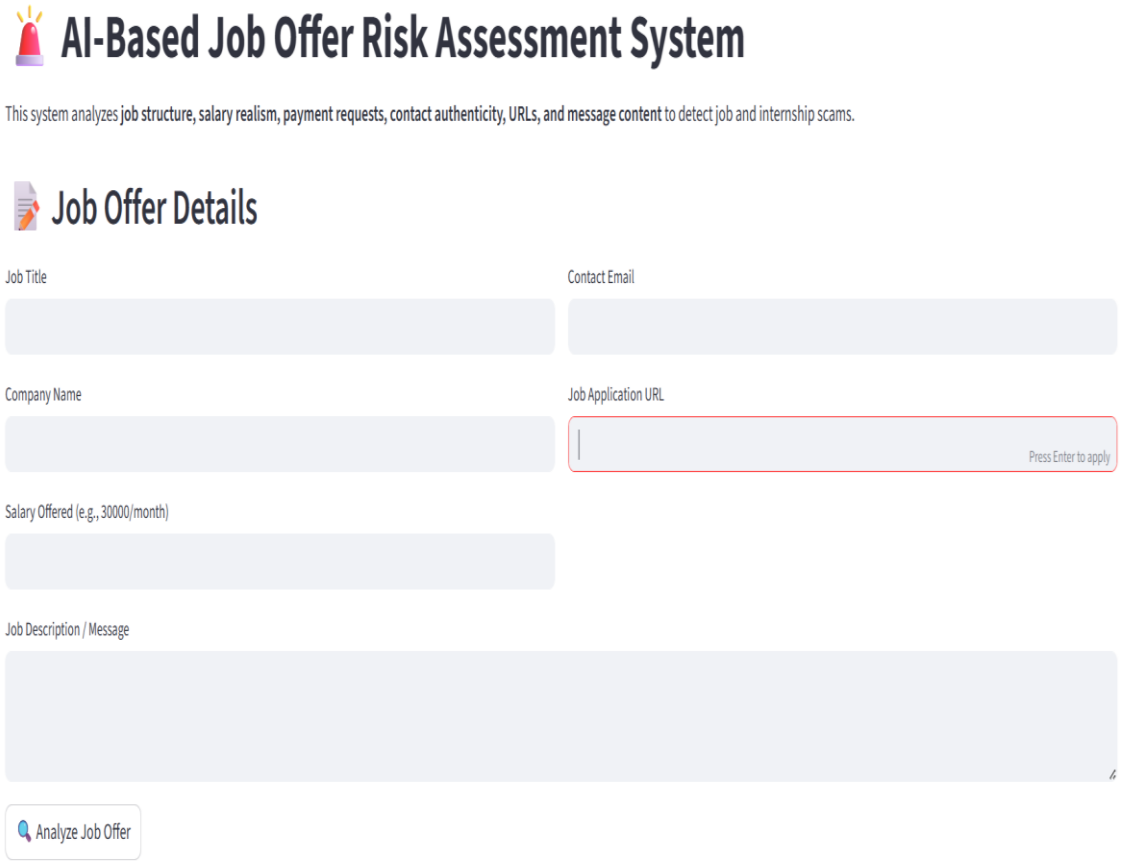
- **TF-IDF Vectorization** – Converts text into numerical form
- **Machine Learning Classifier** – Detects scam-like language
- **Rule-Based Risk Analysis** – Checks salary, payment, email, and URLs
- **Risk Scoring Algorithm** – Calculates final scam risk percentage

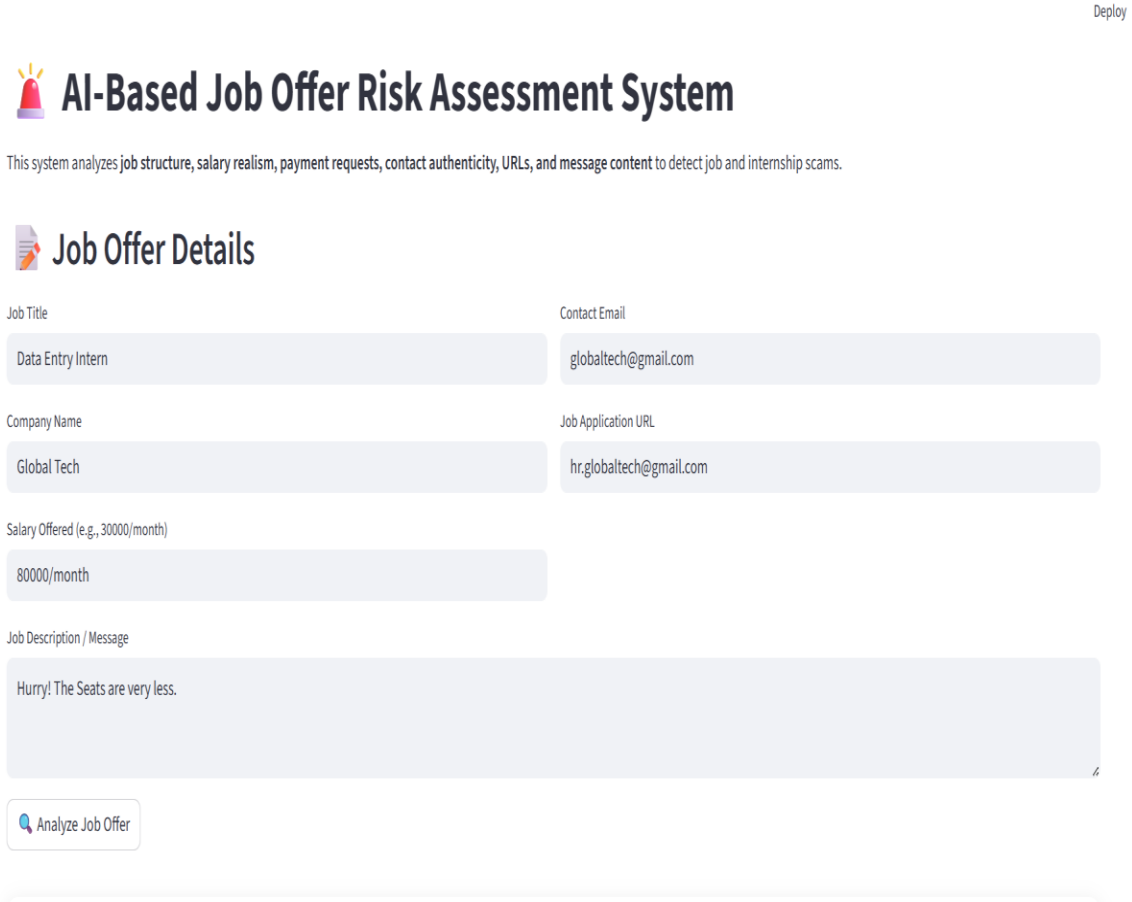
SYSTEM ARCHITECTURE



How the System Works

OUTPUT SCREENS





Analysis Result

Risk Meter

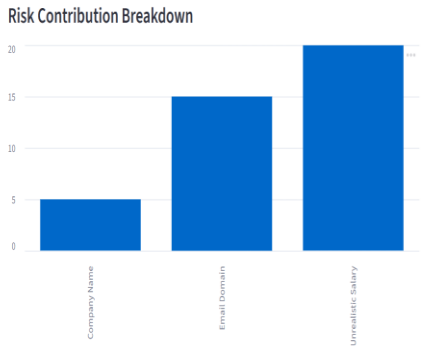
⚠️ **MEDIUM RISK (40%)**

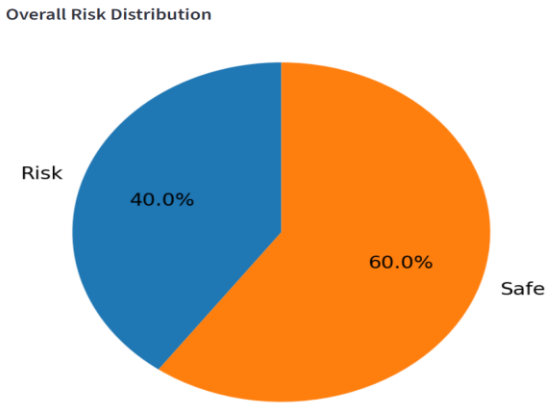
🧠 **Advice:** Verify company details carefully before applying.

Detected Risk Factors

- Unrealistic salary for internship
- Non-official email domain
- Generic or unverifiable company name

📊 Visual Risk Analysis





CONCLUSION

In conclusion, online job and internship scams have become a serious issue affecting students and job seekers, making it essential to have a reliable and automated detection system. The AI-Based Job Offer Risk Assessment and Scam Detection System effectively addresses this problem by using Machine Learning, Natural Language Processing, and rule-based analysis to identify fraudulent job offers. By analyzing job descriptions, salary details, email addresses, and application URLs, the system generates an accurate risk score along with clear visual explanations. This approach not only improves detection accuracy but also helps users understand why a job offer is risky. The system is user-friendly, efficient, and capable of reducing financial loss and data theft. Overall, this project enhances online job safety and provides a practical solution to combat job scams in today’s digital recruitment environment.

Thank you